NovoTech MACHINE TOOLS Inc.

PRODUCT CATALOGUE 2026

CSM · MADE IN TAIWAN · SOLD & SERVICED IN CANADA BY NOVOTECH

HYDRAULIC TUBE BENDERS, END FORMERS AND SAWS

CSM TSR / TDR HYDRAULIC CNC AND TSRJ / TDRJ, TNCPB, TNCB NC PIPE BENDERS TO Ø8-5/8" · F38 / F50 SYMMETRIC BENDERS · EF, REF AND ER END FORMERS · CT325 CIRCULAR SAW







Taiwan
MADE IN TAIWAN

6
MACHINE FAMILIES

1-year
WARRANTY

Canada
PARTS & TECHNICAL SUPPORT

CSM HYDRAULIC TUBE BENDERS

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ABOUT NOVOTECH

YOUR CANADIAN SOURCE FOR CSM HYDRAULIC TUBE BENDERS, END FORMERS AND SAWS

Sales, installation, parts and technical support across Canada.

NovoTech Machine Tools Inc. supplies metalworking and fabrication machinery to job shops, fabricators, maintenance departments and manufacturers across Canada from our base in Mississauga, Ontario. Every machine in this catalogue is sold and supported locally by NovoTech — with spare parts and technical support readily available in Canada.

INCLUDED WITH EVERY NOVOTECH MACHINE

- Made in Taiwan by CSM (Chiao Sheng Machinery) — ISO 9001 and CE
- CSA-approved electrical panel
- 1-year warranty
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  in-stock machines only
- Spare parts & technical support
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- Order today with just 10% down
  0% down financing available
- Installation and training available

CSM · TAINAN, TAIWAN

HYDRAULIC PIPE BENDERS TO Ø8-5/8", AND THE MACHINES AROUND THEM

CSM's hydraulic benders put the force of a hydraulic bend arm, clamp, pressure die and booster on heavy-wall tube and pipe: the TSR / TDR CNC benders with a servo carriage and collet bend pipe to Ø8-5/8" on 20 ft mandrels; the TSRJ / TDRJ and TNCPB are the NC versions with servo or manual feed; the TNCB is the single-axis NC pipe bender to Ø8-5/8"; the F38 / F50 bend both ends of a tube at once. The EF, REF and ER machines form, groove and chamfer the tube ends, and the CT325 saws tube and bundles to Ø4" with servo feed — the stations of a complete tube cell.

Chiao Sheng Machinery (CSM) has built tube benders in Tainan, Taiwan, since 1982 and released its first CNC bender in 1989; the company works from two plants in the Southern Taiwan Science Park, holds 24 international design patents, has been ISO 9001 certified since 2002 and CE-certifies every machine. NovoTech sells, installs and supports CSM machines in Canada from Mississauga, Ontario — layout for your building, installation, operator training, a CSA-approved panel, a 1-year warranty and parts and technical support in Canada.

38 TSR – 220 TSR · TSR-RBF / RBH · 38 TDR – 150 TDR-RBH · 28 MODELS

TSR / TDR HYDRAULIC CNC PIPE BENDERS, Ø1-1/2" TO Ø8-5/8"



MODEL TSR · CSM TSR HYDRAULIC CNC PIPE BENDER



Large pipe elbows bent on the 150 – 220 TSR for shipyard, oil-rig and water-supply work



Furniture and automotive U-bends bent on the 38 – 65 TSR

Hydraulic CNC · Ø1-1/2" – Ø8-5/8" · pipe to 19/32" wall · 1 – 2 tools

The heavy hydraulic CNC bender for pipe and thick-wall tube: a hydraulic bend arm, clamp, pressure die and side booster with a servo carriage and collet rotation, from Ø1-1/2" × 11 ga to Ø8-5/8" × 19/32" pipe on 20 ft mandrels — boiler, construction, shipyard, oil-rig and water-supply pipe. The TDR carries two stacked tools; the -RBF and -RBH versions add a hydraulic rear booster for thin-wall pipe and radii down to 1 × D. Industrial-PC control with 3D simulation, or the PLC control of the NC range.

STANDARD FEATURES

- ✓ Bend arm (C-axis), clamp die, pressure die and side booster by hydraulics
- ✓ Carriage (Y-axis) by servo motor with programmable speed; collet (B-axis) rotation by servo motor
- ✓ Tool / radii shift up-down and left-right by hydraulics (TDR)
- ✓ Mandrel by hydraulics with automatic lubrication and anticipated retraction
- ✓ Hydraulic rear booster on the -RBF / -RBH models
- ✓ Industrial-PC control with touch-screen: bending simulation, 3D preview and interference detection; reads IGS / STP files
- ✓ Programmable speed for every servo axis; USB port for data transfer; manual / automatic mode; inch / metric; XYZ / YBC axes
- ✓ CSA-approved electrical panel; 600 V 3-phase with transformer (208 / 240 / 480 V on request)

OPTIONAL EQUIPMENT

- + Automatic tube loader and unloader; bundle loading
- + Rear booster (servo RBE / hydraulic RBF, RBH) for thin-wall and tight-radius bending
- + Roll-bending head for large radii
- + Integrated circular saw, end-forming station and robot transfer — automated bending cells
- + Tooling: bend dies, clamp dies, pressure dies, wiper dies and mandrels to your tube and radius
- + Custom voltage: 208 V, 240 V, 480 V or 600 V 3-phase

TECHNICAL SPECIFICATIONS · 1 / 2

TSR / TDR HYDRAULIC CNC PIPE BENDERS, Ø1-1/2" TO Ø8-5/8"

TSR

Hydraulic CNC benders, one tool

Model	38 TSR	50 TSR	65 TSR	80 TSR	100 TSR	120 TSR	150 TSR	180 TSR	220 TSR	80 TSR-RBF	100 TSR-RBF	120 TSR-RBF	50 TSR-RBH	65 TSR-RBH	80 TSR-RBH	100 TSR-RBH	120 TSR-RBH	150 TSR-RBH
Max. tube capacity OD × wall, mild steel	Ø1-1/2" (38.1 mm) × 11 ga (3 mm)	Ø2" (50.8 mm) × 11 ga (3 mm)	Ø2-1/2" (65 mm) × 11 ga (3 mm)	Ø3-1/8" (80 mm) × 3/16" (4.5 mm)	Ø4" (101.6 mm) × 7/32" (5.5 mm)	Ø4-3/4" (120 mm) × 7/32" (5.5 mm)	Ø6" (152.4 mm) × 9/32" (7 mm)	Ø7" (180 mm) × 11/32" (9 mm)	Ø8-5/8" (220 mm) × 19/32" (15 mm)	Ø3-1/8" (80 mm) × 7/32" (5.5 mm)	Ø4" (101.6 mm) × 7/32" (5.5 mm)	Ø4-3/4" (120 mm) × 7/32" (5.5 mm)	Ø2" (50.8 mm) × 11 ga (3 mm)	Ø2-1/2" (65 mm) × 11 ga (3 mm)	Ø3-1/8" (80 mm) × 3/16" (4.5 mm)	Ø4" (101.6 mm) × 7/32" (5.5 mm)	Ø4-3/4" (120 mm) × 7/32" (5.5 mm)	Ø6" (152.4 mm) × 9/32" (7 mm)
Centreline radius (CLR)	1.0 – 5.9" (25 – 150 mm)	1.2 – 8.7" (30 – 220 mm)	1.4 – 9.8" (35 – 250 mm)	1.6 – 11.0" (40 – 280 mm)	2.4 – 15.7" (60 – 400 mm)	3.1 – 17.7" (80 – 450 mm)	3.5 – 19.7" (90 – 500 mm)	3.9 – 23.6" (100 – 600 mm)	3.9 – 31.5" (100 – 800 mm)	1.3 – 13.6" (32 – 345 mm)	1.2 – 15.7" (30 – 400 mm)	3.1 – 17.7" (80 – 450 mm)	1.2 – 8.7" (30 – 220 mm)	1.4 – 9.8" (35 – 250 mm)	1.6 – 9.8" (40 – 250 mm)	2.4 – 15.7" (60 – 400 mm)	3.1 – 17.7" (80 – 450 mm)	3.5 – 19.7" (90 – 500 mm)
Mandrel working length	7.2 ft · 87" (2,200 mm)	7.2 ft · 87" (2,200 mm)	8.2 ft · 98" (2,500 mm)	9.8 ft · 118" (3,000 mm)	11.8 ft · 142" (3,600 mm)	16.4 ft · 197" (5,000 mm)	19.7 ft · 236" (6,000 mm)	19.7 ft · 236" (6,000 mm)	19.7 ft · 236" (6,000 mm)	9.8 ft · 118" (3,000 mm)	11.8 ft · 142" (3,600 mm)	16.4 ft · 197" (5,000 mm)	7.2 ft · 87" (2,200 mm)	8.2 ft · 98" (2,500 mm)	9.8 ft · 118" (3,000 mm)	11.8 ft · 142" (3,600 mm)	16.4 ft · 197" (5,000 mm)	19.7 ft · 236" (6,000 mm)
Tooling capacity stacked dies / radii	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Roll bending	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
Rear booster	–	–	–	–	–	–	–	–	–	Hydraulic	Hydraulic	Hydraulic	Hydraulic	Hydraulic	Hydraulic	Hydraulic	Hydraulic	Hydraulic

TECHNICAL SPECIFICATIONS · 2 / 2

TSR / TDR HYDRAULIC CNC PIPE BENDERS, Ø1-1/2" TO Ø8-5/8"

TDR

Hydraulic CNC benders, two stacked tools

Model	38 TDR	50 TDR	65 TDR	80 TDR	100 TDR	120 TDR	150 TDR	100 TDR-RBH	150 TDR-RBH
Max. tube capacity OD × wall, mild steel	Ø1-1/2" (38.1 mm) × 11 ga (3 mm)	Ø2" (50.8 mm) × 11 ga (3 mm)	Ø2-1/2" (65 mm) × 11 ga (3 mm)	Ø3-1/8" (80 mm) × 3/16" (4.5 mm)	Ø4" (101.6 mm) × 7/32" (5.5 mm)	Ø4-3/4" (120 mm) × 7/32" (5.5 mm)	Ø6" (152.4 mm) × 9/32" (7 mm)	Ø4" (101.6 mm) × 7/32" (5.5 mm)	Ø6" (152.4 mm) × 9/32" (7 mm)
Centreline radius (CLR)	1.0 – 5.9" (25 – 150 mm)	1.2 – 8.7" (30 – 220 mm)	1.4 – 9.8" (35 – 250 mm)	1.6 – 11.0" (40 – 280 mm)	2.4 – 15.7" (60 – 400 mm)	3.1 – 15.7" (80 – 400 mm)	3.5 – 19.7" (90 – 500 mm)	2.4 – 15.7" (60 – 400 mm)	3.5 – 19.7" (90 – 500 mm)
Mandrel working length	7.2 ft · 87" (2,200 mm)	7.2 ft · 87" (2,200 mm)	8.2 ft · 98" (2,500 mm)	9.8 ft · 118" (3,000 mm)	11.8 ft · 142" (3,600 mm)	16.4 ft · 197" (5,000 mm)	19.7 ft · 236" (6,000 mm)	11.8 ft · 142" (3,600 mm)	19.7 ft · 236" (6,000 mm)
Tooling capacity stacked dies / radii	2	2	2	2	2	2	2	2	2
Roll bending	–	–	–	–	–	–	–	–	–
Rear booster	–	–	–	–	–	–	–	Hydraulic	Hydraulic

Capacities are the manufacturer's ratings for round mild-steel tube, outside diameter × wall; CLR = centreline radius of the bend die. Mandrel length is the maximum tube length fed with the standard mandrel rod (the working length). Tooling capacity is the number of stacked bend dies / radii; "roll bending" models carry a roll-bending head for large radii; a rear booster (RBE servo, RBF / RBH / CBF hydraulic) pushes the tube from behind for thin-wall and tight-radius bends. Three-phase machines are supplied for 600 V with a CSA-approved panel.

Custom voltage available: 208 V, 240 V, 480 V or 600 V 3-phase.

INCLUDED WITH EVERY NOVOTECH MACHINE

- ✓ Made in Taiwan by CSM (Chiao Sheng Machinery) — ISO 9001 and CE
- ✓ CSA-approved electrical panel
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  0% down financing available
- ✓ Installation and training available

38 – 80 TSRJ · 38 – 65 TDRJ · 50 – 100 TNCPB · 11 MODELS

TSRJ / TDRJ AND TNCPB NC HYDRAULIC TUBE BENDERS, Ø1-1/2" TO Ø4"



MODEL TSRJ · CSM TSRJ NC HYDRAULIC TUBE BENDER WITH SERVO CARRIAGE



CSM 100 TNCPB NC hydraulic tube bender with manual feeding



Servo carriage of the TSRJ / TDRJ

NC hydraulic · Ø1-1/2" – Ø4" · servo feed (TSRJ / TDRJ) or manual feed (TNCPB)

The NC hydraulic benders with a PLC touch-screen: the TSRJ (one tool) and TDRJ (two tools) feed the tube with a servo carriage at programmable speed and rotate it with a servo collet, so multi-plane parts are bent from a stored program; the TNCPB is fed by hand against the stop with servo rotation — the lowest-priced programmed bender for Ø2" to Ø4" furniture, railing and automotive tube. Hydraulic bend arm, clamp, pressure die and side booster; mandrel with automatic lubrication and anticipated retraction.

STANDARD FEATURES

- ✓ Bend arm (C-axis), clamp die, pressure die and side booster by hydraulics
- ✓ Carriage (Y-axis) by servo motor with programmable speed (TSRJ / TDRJ) — manual feeding on the TNCPB
- ✓ Collet (B-axis) grip by hydraulics, rotation by servo motor
- ✓ Mandrel by hydraulics with automatic lubrication and anticipated retraction
- ✓ Tool / radii shift up-down and left-right by hydraulics (TDRJ)
- ✓ PLC control with touch-screen: programmable servo speeds, USB port, manual / automatic, inch / metric; XYZ / YBC axes optional
- ✓ CSA-approved electrical panel; 600 V 3-phase with transformer (208 / 240 / 480 V on request)

OPTIONAL EQUIPMENT

- + Automatic tube loader and unloader; bundle loading
- + Rear booster (servo RBE / hydraulic RBF, RBH) for thin-wall and tight-radius bending
- + Roll-bending head for large radii
- + Integrated circular saw, end-forming station and robot transfer — automated bending cells
- + Tooling: bend dies, clamp dies, pressure dies, wiper dies and mandrels to your tube and radius
- + Custom voltage: 208 V, 240 V, 480 V or 600 V 3-phase

TSRJ / TDRJ — TECHNICAL SPECIFICATIONS

Model	38 TSRJ	50 TSRJ	65 TSRJ	80 TSRJ	38 TDRJ	50 TDRJ	65 TDRJ
Max. tube capacity OD × wall, mild steel	Ø1-1/2" (38.1 mm) × 11 ga (3 mm)	Ø2" (50.8 mm) × 11 ga (3 mm)	Ø2-1/2" (65 mm) × 11 ga (3 mm)	Ø3-1/8" (80 mm) × 3/16" (4.5 mm)	Ø1-1/2" (38.1 mm) × 11 ga (3 mm)	Ø2" (50.8 mm) × 11 ga (3 mm)	Ø2-1/2" (65 mm) × 11 ga (3 mm)
Centreline radius (CLR)	1.0 – 5.9" (25 – 150 mm)	1.2 – 8.7" (30 – 220 mm)	1.4 – 9.8" (35 – 250 mm)	1.6 – 11.0" (40 – 280 mm)	1.0 – 5.9" (25 – 150 mm)	1.2 – 8.7" (30 – 220 mm)	1.4 – 9.8" (35 – 250 mm)
Mandrel working length	7.2 ft · 87" (2,200 mm)	7.2 ft · 87" (2,200 mm)	8.2 ft · 98" (2,500 mm)	9.8 ft · 118" (3,000 mm)	7.2 ft · 87" (2,200 mm)	7.2 ft · 87" (2,200 mm)	8.2 ft · 98" (2,500 mm)
Tooling capacity stacked dies / radii	1	1	1	1	2	2	2
Roll bending	–	–	–	–	–	–	–
Rear booster	–	–	–	–	–	–	–

TECHNICAL SPECIFICATIONS

TSRJ / TDRJ AND TNCPB NC HYDRAULIC TUBE BENDERS, Ø1-1/2" TO Ø4"

TNCPB

NC hydraulic benders with manual feeding

Model	50 TNCPB	75 TNCPB	89 TNCPB	100 TNCPB
Max. tube capacity OD × wall, mild steel	**Ø2" (50.8 mm) × 11 ga (3 mm)**	**Ø3" (76.2 mm) × 11 ga (3 mm)**	**Ø3-1/2" (88.9 mm) × 12 ga (2.5 mm)**	**Ø4" (101.6 mm) × 7/32" (5.5 mm)**
Centreline radius (CLR)	**1.2 – 8.7" (30 – 220 mm)**	**1.6 – 9.8" (40 – 250 mm)**	**2.0 – 11.8" (50 – 300 mm)**	**2.4 – 15.7" (60 – 400 mm)**
Mandrel working length	**6.9 ft · 83" (2,100 mm)**	**7.9 ft · 94" (2,400 mm)**	**8.5 ft · 102" (2,600 mm)**	**11.8 ft · 142" (3,600 mm)**
Tooling capacity stacked dies / radii	1	1	1	1
Roll bending	—	—	—	—
Rear booster	—	—	—	—

Capacities are the manufacturer's ratings for round mild-steel tube, outside diameter × wall; CLR = centreline radius of the bend die. Mandrel length is the maximum tube length fed with the standard mandrel rod (the working length). Tooling capacity is the number of stacked bend dies / radii; "roll bending" models carry a roll-bending head for large radii; a rear booster (RBE servo, RBF / RBH / CBF hydraulic) pushes the tube from behind for thin-wall and tight-radius bends. Three-phase machines are supplied for 600 V with a CSA-approved panel.

Custom voltage available: 208 V, 240 V, 480 V or 600 V 3-phase.

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- ✓ Installation and training available

38 TNCB – 220 TNCB · TNCB-CBF / RBF · 13 MODELS

TNCB SINGLE-AXIS NC HYDRAULIC PIPE BENDERS, Ø1-1/2" TO Ø8-5/8"



MODEL TNCB · CSM TNCB SINGLE-AXIS NC HYDRAULIC PIPE BENDER



Large pipe elbows bent on the 150 – 220 TNCB



Exhaust pipes with flex sections bent on the 38 – 50 TNCB

Single-axis NC hydraulic · Ø1-1/2" – Ø8-5/8" · mandrel · mechanical stop

The single-axis hydraulic bender with a mandrel: the operator positions the pipe against the mechanical stop, the PLC bends it to the programmed angle — the simple, rugged machine for boiler, construction, shipyard and water-supply pipe from Ø1-1/2" to Ø8-5/8", and for furniture and automotive tube on the small models. The -CBF and -RBF versions add a hydraulic booster for thin-wall pipe and tight radii, the 220 TNCB-RBF bends to a 39" centreline radius.

STANDARD FEATURES

- ✓ Bend arm (C-axis), clamp die, pressure die and side booster by hydraulics
- ✓ Mandrel by hydraulics with automatic lubrication and anticipated retraction
- ✓ Mechanical stopper for the feed position
- ✓ Hydraulic booster on the -CBF / -RBF models
- ✓ PLC control with touch-screen: programmable speeds, USB port, manual / automatic, inch / metric
- ✓ CSA-approved electrical panel; 600 V 3-phase with transformer (208 / 240 / 480 V on request)

OPTIONAL EQUIPMENT

- + Automatic loader
- + Hydraulic rear booster
- + Tooling: bend dies, clamp dies, pressure dies, wiper dies and mandrels to your pipe and radius
- + Custom voltage: 208 V, 240 V, 480 V or 600 V 3-phase

TECHNICAL SPECIFICATIONS

TNCB SINGLE-AXIS NC HYDRAULIC PIPE BENDERS, Ø1-1/2" TO Ø8-5/8"

TNCB

Single-axis NC hydraulic benders

Model	38 TNCB	50 TNCB	75 TNCB	89 TNCB	100 TNCB	120 TNCB	150 TNCB	180 TNCB	220 TNCB	75 TNCB-CBF	120 TNCB-CBF	120 TNCB-RBF	220 TNCB-RBF
Max. tube capacity OD × wall, mild steel	Ø1-1/2" (38.1 mm) × 11 ga (3 mm)	Ø2" (50.8 mm) × 11 ga (3 mm)	Ø3" (76.2 mm) × 11 ga (3 mm)	Ø3-1/2" (88.9 mm) × 12 ga (2.5 mm)	Ø4" (101.6 mm) × 7/32" (5.5 mm)	Ø5" (127 mm) × 10 ga (3.5 mm)	Ø6" (152.4 mm) × 9/32" (7 mm)	Ø7" (180 mm) × 11/32" (9 mm)	Ø8-5/8" (220 mm) × 5/16" (8 mm)	Ø3" (76.2 mm) × 11 ga (3 mm)	Ø5" (127 mm) × 7/32" (5.5 mm)	Ø5" (127 mm) × 7/32" (5.5 mm)	Ø8-5/8" (220 mm) × 9/32" (7 mm)
Centreline radius (CLR)	1.0 – 6.7" (25 – 170 mm)	1.2 – 9.8" (30 – 250 mm)	1.6 – 11.8" (40 – 300 mm)	1.6 – 13.8" (40 – 350 mm)	2.4 – 15.7" (60 – 400 mm)	3.1 – 17.7" (80 – 450 mm)	3.5 – 19.7" (90 – 500 mm)	3.9 – 23.6" (100 – 600 mm)	3.9 – 27.6" (100 – 700 mm)	1.6 – 11.8" (40 – 300 mm)	3.1 – 15.7" (80 – 400 mm)	2.4 – 17.7" (60 – 450 mm)	3.9 – 39.4" (100 – 1000 mm)
Mandrel working length	6.2 ft · 75" (1,900 mm)	7.2 ft · 87" (2,200 mm)	8.5 ft · 102" (2,600 mm)	8.5 ft · 102" (2,600 mm)	11.8 ft · 142" (3,600 mm)	17.1 ft · 205" (5,200 mm)	19.7 ft · 236" (6,000 mm)	19.7 ft · 236" (6,000 mm)	19.7 ft · 236" (6,000 mm)	8.5 ft · 102" (2,600 mm)	17.1 ft · 205" (5,200 mm)	17.1 ft · 205" (5,200 mm)	19.7 ft · 236" (6,000 mm)
Tooling capacity stacked dies / radii	1	1	1	1	1	1	1	1	1	1	1	1	1
Roll bending	–	–	–	–	–	–	–	–	–	–	–	–	–
Rear booster	–	–	–	–	–	–	–	–	–	Hydraulic	Hydraulic	Hydraulic	Hydraulic

Capacities are the manufacturer's ratings for round mild-steel tube, outside diameter × wall; CLR = centreline radius of the bend die. Mandrel length is the maximum tube length fed with the standard mandrel rod (the working length). Tooling capacity is the number of stacked bend dies / radii; "roll bending" models carry a roll-bending head for large radii; a rear booster (RBE servo, RBF / RBH / CBF hydraulic) pushes the tube from behind for thin-wall and tight-radius bends. Three-phase machines are supplied for 600 V with a CSA-approved panel.

Custom voltage available: 208 V, 240 V, 480 V or 600 V 3-phase.

INCLUDED WITH EVERY NOVOTECH MACHINE

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  0% down financing available
- ✓ Installation and training available

F38 · F50 · F38-90 · F38H-4A · F38H-5A · 5 MODELS

F38 / F50 SYMMETRIC TUBE BENDERS — BOTH ENDS OF THE TUBE BENT AT ONCE, Ø1-1/2" AND Ø2"



MODEL F38 · CSM F38H-5A MULTI-AXIS SYMMETRIC TUBE BENDER WITH SERVO FEEDING



F38 single-axis symmetric bender — two heads bend both ends of the tube at once

Two heads · Ø 1-1/2" and Ø 2" × 12 ga · 1 to 3 axes · stacked tools

Two bending heads on one bed bend both ends of a tube in the same stroke — towel rails, handles, furniture frames, U-bolts, chair legs and headboards come off the machine finished, at twice the output of a single-head bender. The F38 and F50 are the single-axis hydraulic machines with a head pusher for fast unloading; the F38-90 adds manual multi-plane rotation; the F38H-4A and F38H-5A feed the tube with a servo motor and the -5A rotates it with a servo collet for automatic multi-plane parts. Stacked tools for Ø3/4", Ø1", Ø1-1/4" and Ø1-1/2".

STANDARD FEATURES

- ✓ Two bending heads bend both tube ends in one cycle; bending, clamp die and pressure die by hydraulics
- ✓ Head pusher for fast unloading (single-axis models)
- ✓ Feeding by servo motor (F38H-4A / 5A); grip by hydraulics with servo rotation for multi-plane bends (F38H-5A)
- ✓ Stacked tooling: up to 3 – 4 dies for Ø3/4", 2 – 3 for Ø1", 1 – 2 for Ø1-1/4", 1 for Ø1-1/2"
- ✓ CSA-approved electrical panel; 600 V 3-phase with transformer (208 / 240 / 480 V on request)

OPTIONAL EQUIPMENT

- + Tooling to your tube and radius
- + Custom voltage: 208 V, 240 V, 480 V or 600 V 3-phase

F SERIES — TECHNICAL SPECIFICATIONS

Model	F38	F50	F38-90	F38H-4A	F38H-5A
Max. tube capacity OD × wall, mild steel	**Ø1-1/2" × 12 ga (38.1 × 2.5 mm)**	**Ø2" × 14 ga (50 × 2 mm)**	**Ø1-1/2" × 12 ga (38.1 × 2.5 mm)**	**Ø1-1/2" × 12 ga (38 × 2.5 mm)**	**Ø1-1/2" × 12 ga (38 × 2.5 mm)**
Centreline radius (CLR)	**1.5 – 7.1" (38 – 180 mm)**	**1.6 – 9.8" (40 – 250 mm)**	**1.5 – 7.1" (38 – 180 mm)**	**1.5 – 7.1" (38 – 180 mm)**	**1.5 – 7.1" (38 – 180 mm)**
Working distance between the two bends	**6.7 – 59.1" (170 – 1500 mm)**	**7.9 – 90.6" (200 – 2300 mm)**	**6.7 – 59.1" (170 – 1500 mm)**	**9.8 – 165.4" (250 – 4200 mm)**	**12.6 – 165.4" (320 – 4200 mm)**
Tools stack Ø3/4" / Ø1" / Ø1-1/4" / Ø1-1/2"	3-4 / 2-3 / 1-2 / 1	4-5 / 3-4 / 2-3 / 1-2	3-4 / 2-3 / 1-2 / 1	3-4 / 2-3 / 1-2 / 1	3-4 / 2-3 / 1-2 / 1
Multi-plane	—	—	Manual	—	Automatic
Axes	1	1	1	2	3

Capacities for round mild-steel tube; the working distance is the range of centre-to-centre spacing between the two bends; tools stack = number of dies stacked for that tube size.

Custom voltage available: 208 V, 240 V, 480 V or 600 V 3-phase.

EF20 · EF30 · EF50 · EF20VH – EF50VH · REF30 · REF40 · ER30-5 · 9 MODELS

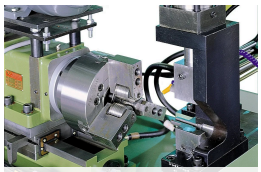
EF SIX-STATION TUBE END FORMERS AND REF / ER GROOVING AND END-FINISHING MACHINES, Ø13/16" TO Ø2"



MODEL EF · CSM EF 30-6 SIX-STATION TUBE END FORMING MACHINE



CSM REF 40 tube grooving machine



Rotary grooving head of the CSM REF machine

Six forming stations · Ø 13/16" – Ø 2" · grooving to Ø 1-9/16" · end forming + chamfering + grooving

CSM's end-finishing machines for the tube that comes off its benders: the EF series carries six forming stations on one hydraulic ram — expand, reduce, bead, flare and flange in sequence without a tool change — for tube to Ø13/16", Ø1-3/16" and Ø2", in a horizontal (plane) or vertical (VH) layout; the REF 30 and REF 40 roll a groove into the tube end for O-ring and clamp connections; the ER 30-5 does end forming, chamfering and grooving on one machine — the complex end-finish station of an automated bending cell. PLC control with touch-screen, hydraulic clamping, servo feeding on the ER.

STANDARD FEATURES

- ✓ EF: six forming stations on one ram — expand, reduce, bead, flare, flange without a tool change; plane (horizontal) or VH (vertical) layout
- ✓ REF: rotary grooving for O-ring and clamp grooves, 3/16" – 1/4" forming stroke
- ✓ ER 30-5: end forming, chamfering and grooving in one cycle; trolley platform for moving between cells
- ✓ Hydraulic clamping and forming; PLC with touch-screen; working height 39 – 49"
- ✓ CSA-approved electrical panel; 600 V 3-phase with transformer (208 / 240 / 480 V on request)

OPTIONAL EQUIPMENT

- + Forming tools to your tube end
- + Integration into a bending cell with robot transfer
- + Custom voltage: 208 V, 240 V, 480 V or 600 V 3-phase

EF — TECHNICAL SPECIFICATIONS

Model	EF20	EF30	EF50	EF20VH	EF30VH	EF50VH
Max. tube capacity OD	**Ø13/16" (20 mm)**	**Ø1-3/16" (30 mm)**	**Ø2" (50 mm)**	**Ø13/16" (20 mm)**	**Ø1-3/16" (30 mm)**	**Ø2" (50 mm)**
Forming stations	**6**	**6**	**6**	**6**	**6**	**6**
Layout	Horizontal	Horizontal	Horizontal	Vertical	Vertical	Vertical
Working height	48.0" (1,220 mm)	48.0" (1,220 mm)	49.2" (1,250 mm)	48.0" (1,220 mm)	48.0" (1,220 mm)	49.2" (1,250 mm)
Total power	7.4 HP (5.5 kW)	7.4 HP (5.5 kW)	10.1 HP (7.5 kW)	7.4 HP (5.5 kW)	7.4 HP (5.5 kW)	10.1 HP (7.5 kW)
Max. pressure	2,418 psi (170 kgf/cm^2)	2,418 psi (170 kgf/cm^2)	2,418 psi (170 kgf/cm^2)	1,991 psi (140 kgf/cm^2)	1,991 psi (140 kgf/cm^2)	2,418 psi (170 kgf/cm^2)

TECHNICAL SPECIFICATIONS

EF SIX-STATION TUBE END FORMERS AND REF / ER GROOVING AND END-FINISHING MACHINES, Ø13/16" TO Ø2"

REF / ER

Grooving and complex end-finishing

Model	REF30	REF40	ER30-5
Max. tube capacity OD	**Ø1-3/16" (30 mm)**	**Ø1-9/16" (40 mm)**	**Ø1-3/16" (30 mm)**
Max. forming stroke	**1/4" (6.5 mm)**	**3/16" (5 mm)**	**1/4" (6.5 mm)**
Operations	Grooving	Grooving	End forming, chamfering, grooving
Working height	41.7" (1,060 mm)	47.6" (1,210 mm)	39.4" (1,000 mm)
Total power	5.0 HP (3.7 kW)	5.9 HP (4.4 kW)	7.4 HP (5.5 kW)
Max. pressure	1,422 psi (100 kgf/cm^2)	996 psi (70 kgf/cm^2)	2,418 psi (170 kgf/cm^2)

Capacities are the maximum tube outside diameter; forming stroke on the REF / ER is the depth of the groove or form.

Custom voltage available: 208 V, 240 V, 480 V or 600 V 3-phase.

INCLUDED WITH EVERY NOVOTECH MACHINE

- ✓ Made in Taiwan by CSM (Chiao Sheng Machinery) — ISO 9001 and CE
- ✓ CSA-approved electrical panel
- ✓ 1-year warranty
- ✓ 7-day money-back guarantee
  in-stock machines only
- ✓ Spare parts & technical support
  readily available in Canada
- ✓ Order today with just 10% down
  0% down financing available
- ✓ Installation and training available

CT325 L · M · H · 3 MODELS

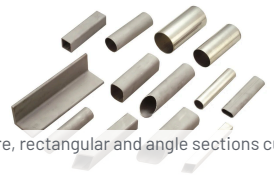
CT325 HYDRAULIC CIRCULAR SAWS WITH SERVO FEED, Ø1/2" TO Ø4" TUBE AND BUNDLES



MODEL CT325 · CT325 HYDRAULIC CIRCULAR SAW WITH SERVO FEED, PLC AND PEDESTAL CONTROL



Bundle-cutting vise of the CT325 — nine tubes cut at once



Round, square, rectangular and angle sections cut on the CT325

Ø1/2" – Ø4" · bundle cutting · servo feed ±0.004" · 3 speed ranges

The automatic circular saw CSM builds for its bending cells: hydraulic down-feed of a Ø10 – 12-3/4" blade driven through planetary gears, hydraulic clamping, a servo carriage that feeds the tube to ±0.004" and a PLC with touch-screen — round, square, rectangular and angle sections, single or in bundles with the customised vises. Three spindle-speed ranges for steel (L), stainless and alloy (M) and aluminum and non-ferrous (H), all at 3.8 HP.

STANDARD FEATURES

- ✓ Hydraulic down-feed; blade driven through planetary transmission gears for low energy use
- ✓ Hydraulic clamping; feeding by servo motor to ±0.004" at 39 in/s
- ✓ Standard vises for Ø1-3/4" – Ø4"; customised vises for Ø1/2" – Ø2-1/2" and bundle cutting
- ✓ PLC control with touch-screen; standing pedestal with emergency stop and remote control
- ✓ CSA-approved electrical panel; 600 V 3-phase with transformer (208 / 240 / 480 V on request)

OPTIONAL EQUIPMENT

- + Bundle-cutting vises
- + Saw blades for steel, stainless and aluminum
- + Integration into a bending cell
- + Custom voltage: 208 V, 240 V, 480 V or 600 V 3-phase

CT325 — TECHNICAL SPECIFICATIONS

Model	CT325 L	CT325 M	CT325 H
Capacity, mild steel standard vises / customised vises	**Ø1-3/4" – 4" (44 – 101.6 mm) / Ø1/2" – 2-1/2" (12 – 63.5 mm)**	**Ø1-3/4" – 4" (44 – 101.6 mm) / Ø1/2" – 2-1/2" (12 – 63.5 mm)**	**Ø1-3/4" – 4" (44 – 101.6 mm) / Ø1/2" – 2-1/2" (12 – 63.5 mm)**
Saw blade	**Ø9-7/8" – 12-3/4" (250 – 325 mm)**	**Ø9-7/8" – 12-3/4" (250 – 325 mm)**	**Ø9-7/8" – 12-3/4" (250 – 325 mm)**
Spindle speed	**22 – 62 rpm**	**47 – 135 rpm**	**80 – 220 rpm**
Total power	3.8 HP (2.8 kW)	3.8 HP (2.8 kW)	3.8 HP (2.8 kW)
Feeding speed / accuracy	39 in/s (1,000 mm/s) / ±0.004" (±0.1 mm)	39 in/s (1,000 mm/s) / ±0.004" (±0.1 mm)	39 in/s (1,000 mm/s) / ±0.004" (±0.1 mm)
Working height	37.4" (950 mm)	37.4" (950 mm)	37.4" (950 mm)
Dimensions L × W × H	67 × 30 × 63" (1,700 × 760 × 1,600 mm)	67 × 30 × 63" (1,700 × 760 × 1,600 mm)	67 × 30 × 63" (1,700 × 760 × 1,600 mm)
Weight	1,764 lb (800 kg)	1,764 lb (800 kg)	1,764 lb (800 kg)

Capacities for mild steel (40 kgf/mm^2); the three types differ in spindle speed range only.

Custom voltage available: 208 V, 240 V, 480 V or 600 V 3-phase.

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Specifications subject to change without notice. Imperial values are converted from manufacturer metric data and rounded: thin sheet in gauge (ga), plate and capacities in common fractions (1/2, 1/4, 1/8, 1/16). Model numbers are manufacturer designations. Photos may show optional equipment. 7-day money-back guarantee on in-stock machines only; financing available.